



RELIABILITY IN FLIGHT WITH PRECISION IN HEAT MANAGEMENT THERMAL DESIGN OF AIRBORNE ELECTRONICS



Figure — Generic Avionics LRU Enclosure (illustrative)

Shape Your Idea To Reality
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Avionics LRU & Our Solutions

In avionics, Line Replacement Units (LRUs) are the backbone of mission-critical systems. But as aircraft electronics grow more compact and powerful, overheat becomes the hidden risk — threatening reliability, safety, and compliance.

Our **Thermal Design & Simulation Services** ensure your LRUs operate cooler, last longer, and meet stringent aerospace standards. By combining advanced CFD modelling, virtual prototyping, and optimization workflows, we help you predict heat flow, eliminate hotspots, and design enclosures that withstand the toughest flight conditions.

Our Services

01 · Cooling Design Solutions

- Heat distribution prediction in avionics LRUs
- Identification of hotspots before physical prototyping
- Virtual testing under flight conditions
- Airflow channel design for efficient cooling
- Heat sink and TIM Integration
- Lightweight yet robust enclosure solutions

02 · Cooling Method Trade Studies

Evaluation of conduction-cooled, forced-air, and cold-plate architectures for optimal SWaP-C performance.

03 · Altitude & Environmental Derating

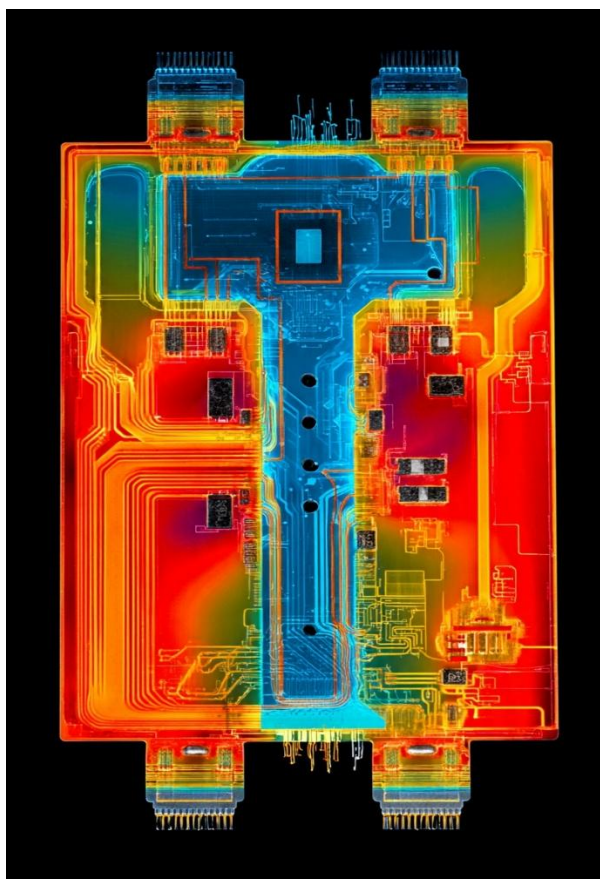
Analysis of reduced convective cooling and pressure effects up to 55,000 ft per operational envelope.

04 · Test Correlation

Chamber and bench test support, including thermal imaging and embedded sensors, to correlate and refine thermal models against measured data.

05 · Compliance Reporting

Documentation aligned with DO-160 qualification for certification.



Technical Capabilities

STUDY PARAMETERS	SIMULATION OUTPUT
Component-level insights	Junction Temperature, Case Temperature
Enclosure Surface	External case/touch temperature at maximum ambient
Operating Temperature Range	Functional range across the flight envelope
Altitude Derating	Reduced air density impact on convective cooling (up to 55,000 ft)
Thermal Cycling Endurance	Repeated cycling fatigue on solder joints/interfaces

Compliance & Certification Standards

Verification against Industry & Regulatory Standard

- RTCA DO-160 (Section 4 & 5) — Environmental Conditions and Test Procedures for Airborne Equipment: Temperature, Altitude, and Temperature Variation.
- MIL-STD-810H — Environmental Engineering Considerations (Method 501.7 High Temp, 502.7 Low Temp, 503.7 Temp Shock).
- MIL-STD-461G — EMI/thermal interface considerations for co-located electronics.
- ARP4761 / ARP4754A — Safety assessment and system development processes referencing thermal failure modes.
- SAE ARP1834 / GJB — Thermal design guidance for airborne electronic equipment (as applicable by program).
- Manufacturer/OEM Interface Control Document (ICD) — enclosure-specific thermal & mounting requirements.

Deliverables

Thermal Analysis Report (TAR) documenting

Simulation Modelling, Inputs, Assumptions & Boundary conditions

Simulation Results: Heat maps, temperature distribution plots, airflow patterns etc.

Simulation dashboards with performance metrics

Executive summary highlighting risks and recommendations

Design Recommendations

Risk Assessment

Compliance-ready documentation for certification

Engagement Model

Requirement Analysis: Scoping workshop to capture the power loss map, duty cycle, and installation zone etc.

Pre-Design/Simulation Review: Enclosure Design, CFD/FE model build, boundary condition definition, and analysis execution.

Pre-Report: Draft TAR issued for customer review, followed by chamber/bench test correlation (if required).

Final certification-ready report delivered with margin summary and compliance mapping.



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